

CSF – blood, paired within donor

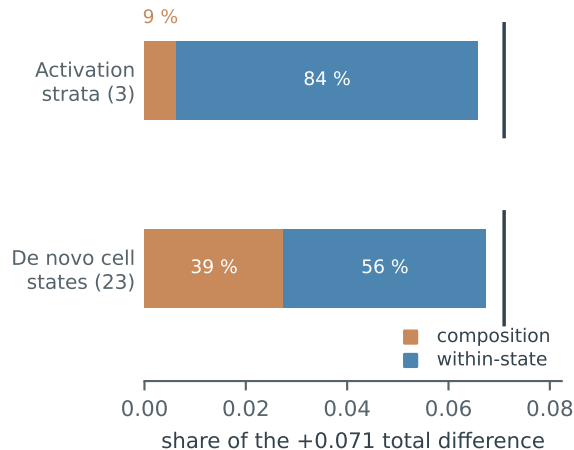
## A The elevation is the same at every activation level

one dot per donor · bar = median · shaded = 95 % of matched random gene sets



## B Part of it is a mixture shift

vertical rule = total · Kitagawa split, exact



GSE138266, 10 donors contributing paired CSF and blood; donor is the unit. Activation is scored from 21 genes verified disjoint from both modules.

In A the activation score itself is flat within strata (empirical p 0.66–0.92), so the strata are activation-matched. The pro-inflammatory arm does not reach BH  $\leq 0.05$  against its matched null (BH 0.063–0.068); act1 draws nine donors because one unit falls below the 50-cell floor.